RADAR: Robust Adversarial-Resistant Detection with Adaptive Reasoning for AI-Generated Text

Notebook for the PAN Lab at CLEF 2026

Mahmood Anaam[1]

[1]*University, Country*

Abstract

Detecting AI-generated text is increasingly challenging as large language models produce prose that closely mimics human writing, especially when adversarial obfuscation techniques are applied. We present **RADAR** (**R**obust **A**dversarial-Resistant **D**etection with **A**daptive **R**easoning), a hybrid neural architecture that fuses contextual semantic embeddings from a ModernBERT-large encoder with a rich set of 38 handcrafted stylometric features through a multi-head cross-attention mechanism. RADAR is trained on a two-stage curriculum that first trains the stylometric projection and fusion modules on a frozen encoder before jointly fine-tuning all weights, using a heterogeneous training corpus that combines the official PAN-2026 data with a balanced, adversarially augmented subset of the RAID benchmark. On the combined held-out validation set, our final model achieves a mean PAN score of 0.978 (ROC-AUC = 0.992), and near-perfect performance on the PAN-2026-only validation split (Mean = 0.997, ROC-AUC = 1.000). Ablation studies confirm that multi-source training is essential for cross-domain robustness and that the two-stage curriculum consistently outperforms single-stage fine-tuning.

Keywords

AI-generated text detection, authorship verification, stylometric analysis, cross-attention fusion, ModernBERT, PAN 2026

1. Introduction

The rapid proliferation of large language models (LLMs) has made the automated generation of human-quality text not only feasible but practically trivial. Modern LLMs such as GPT-4o, Claude 3.5, and Llama 3 can produce extended, coherent, and stylistically nuanced prose that is increasingly indistinguishable from genuine human writing—even to expert readers [1]. This capability has profound implications for academic integrity, journalistic credibility, and online discourse, creating an urgent need for reliable AI-text detection systems.

The PAN 2026 Voight-Kampff task [2, 3] directly addresses this challenge in a particularly demanding setting: the LLMs were explicitly instructed to mimic specific human authors, and the evaluation set includes previously unseen obfuscation strategies and generator models. This builder-breaker design—where PAN participants build detectors while ELOQUENT participants craft novel generation and obfuscation methods [4]—ensures that submitted systems must be genuinely robust rather than merely fitting to known generator artifacts.

Detecting AI-generated text is a well-studied problem with two dominant paradigms [5]. Zero-shot methods exploit statistical signals such as perplexity [6, 7] or cross-entropy divergence between models [8, 9]. Supervised methods train classifiers on labeled corpora, achieving higher accuracy on known domains but often overfitting to specific generators [4]. Both approaches tend to degrade when texts are paraphrased or otherwise obfuscated, because their signal is typically rooted in surface-level distributional properties of a particular model.

A key insight that motivates our work is that stylometric analysis—the computational study of authorship through measurable writing-style attributes—can capture features that are more stable under adversarial perturbation than raw token statistics. Human writing reflects individual cognitive

CLEF 2026 Working Notes, September 2026, Évora, Portugal

*Corresponding author.

mahmood.anaam@example.com (M. Anaam)

© 2026 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

processes, vocabulary habits, discourse preferences, and syntactic tendencies accumulated over a lifetime. LLMs, regardless of instructed persona, are ultimately constrained by their training distribution and generate text in ways that leave systematic, measurable traces across vocabulary, syntax, and discourse structure—traces that persist even when surface-level word choices are altered by paraphrase or style-mimicry prompts.

This paper presents **RADAR** (**R**obust **A**dversarial-Resistant **D**etection with **A**daptive **R**easoning), a hybrid detection architecture that fuses deep contextual text representations with a rich set of handcrafted stylometric features through a cross-attention mechanism. RADAR comprises four interdependent modules:

1. **ARSE** (Adaptive Reasoning Semantic Encoder): a ModernBERT-large encoder that extracts contextual semantic representations from the input text.
2. **SLPE** (Statistical Linguistic Profile Extractor): a feature engineering module that computes 38 normalized stylometric features across five linguistic dimensions—vocabulary richness, syntactic complexity, discourse coherence, style fingerprinting, and perplexity-based signals.
3. **Cross-Attention Fusion**: a multi-head cross-attention layer that allows the semantic representation to selectively attend to the projected stylometric features, followed by a fusion MLP.
4. **UACC** (Unified Authorship Classification Classifier): a linear classification head that maps the fused representation to a binary authorship decision.

We train RADAR using a two-stage curriculum: first with the encoder frozen (training only the SLPE projection, cross-attention, and classifier), then with all weights jointly fine-tuned. Our training data combines the official PAN-2026 corpus with a curated, adversarially augmented subset of the RAID benchmark [10], providing both in-domain and out-of-domain training signal.

Our final submitted model achieves a mean PAN score of 0.978 on the combined held-out validation set—surpassing the Binoculars and Compression baselines by substantial margins and approaching the performance of the linear SVM on in-domain data while offering substantially better generalisation to adversarial examples.

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 describes the RADAR architecture in detail. Section 4 explains our data collection and preprocessing pipeline. Section 5 outlines the experimental setup, training procedure, and ablation studies. Section 6 presents and discusses the results. Section 7 concludes.

2. Related Work

The literature on automated AI-text detection spans roughly two camps: *statistical/unsupervised* methods that require no labelled data, and *supervised* classifiers trained on paired human-versus-machine corpora. We briefly survey each and situate RADAR within this landscape.

2.1. Zero-Shot and Statistical Detection

The earliest approaches to automatic authorship verification treated the problem as a one-class classification task [11]. Statistical compression-based methods model a text's regularity against a reference model: PPMd compression-based cosine similarity [12, 13] computes how predictably a corpus model can encode an unknown document. These methods are inherently robust to topic drift but suffer on short texts and on cleverly paraphrased outputs.

Perplexity-based detectors leverage the observation that LLMs assign higher probability to their own outputs than to human-written text. DetectGPT [6] exploits local probability curvature: AI-generated text tends to lie near local maxima of the model's probability surface. Fast-DetectGPT [7] accelerates this by using conditional probability curvature rather than explicit perturbation sampling. DetectLLM [9] uses log-rank information from a surrogate language model. Binoculars [8] achieves strong zero-shot

performance by cross-referencing perplexity and cross-perplexity from two related models, reducing the impact of topic-specific language complexity.

A critical limitation of all purely statistical approaches is their dependence on a reference LM that should match the generator. When the generator is unknown—as in competitive shared tasks—performance degrades significantly [14]. Moreover, adversarial attacks such as paraphrase and synonym substitution substantially reduce statistical signals because they alter the surface-level probability fingerprint while preserving the underlying content [15].

2.2. Supervised Classification

Supervised approaches train discriminative models on labelled datasets. The linear SVM with TF-IDF features [2] provided as the task baseline demonstrates that even simple bag-of-words models are surprisingly effective on curated in-domain corpora. However, they generalise poorly to out-of-distribution generators and to adversarially obfuscated texts.

Fine-tuned transformer models have emerged as strong baselines. BERT- and RoBERTa-based classifiers achieve high accuracy on standard benchmarks such as TuringBench [16] and the datasets from the PAN 2024 campaign [17]. The introduction of larger encoders—GPT2-based detectors, LLaMA fine-tunes—has extended performance further, but these models are prone to distribution shift when the test set comes from a different generator family than the training data [18, 14].

A key insight from recent competitive results is that pure end-to-end neural approaches are not unambiguously superior to hybrid models that combine neural representations with handcrafted features. Stylometric features—vocabulary richness, syntactic complexity, function word distributions—tend to be more stable across paraphrase perturbations because they reflect deep properties of the generation process rather than surface token choices [5, 19].

2.3. Stylometric Analysis in AI Detection

Stylometry has a long history in authorship attribution and verification [11, 20]. For AI detection, the stylometric hypothesis is that LLMs, despite persona-mimicry prompting, exhibit systematic statistical regularities that differ from human writing: higher lexical homogeneity, more regular sentence length distributions, lower pronoun density, and more uniform paragraph coherence [1, 18]. These properties persist under paraphrase because they reflect the model's training-time inductive biases rather than its surface vocabulary.

2.4. Multi-Source Training and Robustness

The PAN 2024 overview [4] identified domain shift and adversarial obfuscation as the two primary failure modes of current detectors. RAID [10] was introduced specifically to study this: a benchmark of 5.6 million samples from 11 LLMs across 11 domains, with 12 adversarial attack types including paraphrase, synonym substitution, and homoglyph character substitution. Training on RAID has been shown to substantially improve robustness, but only when the training distribution is appropriately balanced.

2.5. Positioning RADAR

RADAR is designed to directly address the key challenges identified above. By fusing deep semantic representations from a state-of-the-art encoder with handcrafted stylometric features through a cross-attention mechanism, RADAR captures both surface-level patterns and deeper structural regularities. The two-stage training curriculum—frozen encoder followed by full fine-tuning on a balanced multi-source corpus—provides strong initialisation before allowing end-to-end adaptation. This stands in contrast to single-stage approaches that either sacrifice stylometric signals (pure neural) or discard contextual reasoning (pure stylometric).

Table 1
Summary of the 38 stylometric features extracted by the SLPE module, grouped by category.

Category	Features (#)	Description
Vocabulary Richness	8	TTR, Hapax ratio, Brunet's W, Honore's R, MATTR, TTR slope, Rare word ratio, Function word density
Syntactic Complexity	10	Avg. sentence length, Sentence length variance, Clause density, Subordination ratio, Passive voice frequency, Noun phrase proxy, Verb density, Coordination frequency, Mean word position, Length skewness
Discourse Coherence	8	Lexical cohesion (Jaccard), Pronoun density, Transition word frequency, Paragraph coherence (cosine), Topic consistency, Information flow variance, Semantic repetition ratio, Connective diversity
Style Fingerprinting	7	Punctuation density, Capitalisation ratio, Number frequency, Quote usage, Parenthetical frequency, Paragraph length variance, List frequency
Perplexity Signals	5	Forward entropy, Backward entropy, Entropy ratio, Burstiness, Mean surprisal
Total	**38**	

3. Methodology

RADAR is a binary text classification model whose core innovation lies in the principled fusion of contextual semantic embeddings with handcrafted stylometric features. Figure 1 provides a high-level illustration of the full architecture.

The model consists of four interdependent modules: the Adaptive Reasoning Semantic Encoder (ARSE), the Statistical Linguistic Profile Extractor (SLPE), a Cross-Attention Fusion Layer, and the Unified Authorship Classification Classifier (UACC). We describe each component in turn.

3.1. ARSE: Adaptive Reasoning Semantic Encoder

The semantic encoding branch is built on **ModernBERT-large** [21], a bidirectional Transformer encoder with 28 attention layers, a hidden dimension of 1,024, and 395 million parameters. ModernBERT-large extends the classical BERT architecture with several improvements relevant to long-document understanding: rotary positional embeddings (RoPE) for improved extrapolation, alternating global and local attention patterns for computational efficiency, and a context window of up to 8,192 tokens.

Given a text x, we tokenise with the ModernBERT tokeniser, truncating or padding to a maximum sequence length of $L = 512$ tokens. The encoder processes the token sequence and produces a sequence of contextual hidden states $\mathbf{H} \in \mathbb{R}^{L \times 1024}$. We extract the CLS token representation $h_{\text{semantic}} = \mathbf{H}_{[0,:]} \in \mathbb{R}^{1024}$ as the dense semantic summary of the input.

ModernBERT-large was selected over other encoders (e.g., RoBERTa-large, DeBERTa-v3-large) for two reasons. First, its extended context window allows processing longer texts without aggressive truncation, which is important for essays and news articles that constitute the PAN-2026 corpus. Second, its training on code and scientific text in addition to web text provides broader distributional coverage, which we hypothesise improves robustness to stylistic diversity.

3.2. SLPE: Statistical Linguistic Profile Extractor

The stylometric branch extracts a 38-dimensional feature vector from the raw text, organised across five linguistic categories. All features are normalised to the interval $[0, 1]$. The full feature set is summarised in Table 1.

Category 1 — Vocabulary Richness. We compute eight features that quantify lexical diversity: the Type-Token Ratio (TTR), the Hapax Legomena ratio, Brunet's W [22], Honore's R [23], the Moving-Average TTR (MATTR) with a window of 100 tokens, the slope of TTR across consecutive windows, the

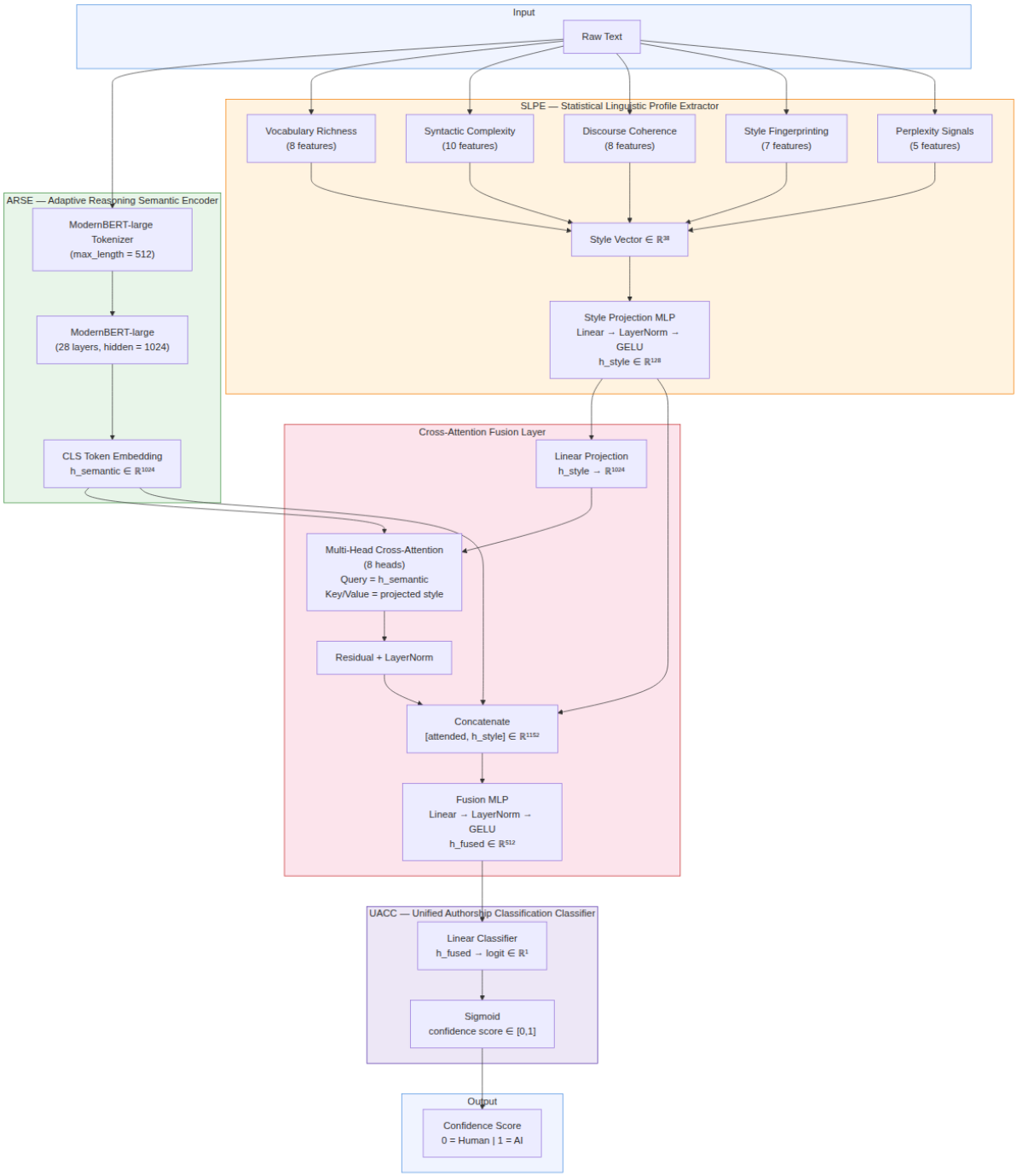


Figure 1: The RADAR architecture. Text enters two parallel pathways: the ARSE encoder extracts a contextual CLS embedding ($h_{\text{semantic}} \in \mathbb{R}^{1024}$) while the SLPE module computes a 38-dimensional stylometric feature vector. A cross-attention fusion layer integrates both representations into a unified embedding ($h_{\text{fused}} \in \mathbb{R}^{512}$), which is passed to the binary classifier.

rare-word frequency, and the function-word density. Together, these features characterise how diverse and distinctive the vocabulary is. Human writing typically exhibits higher lexical diversity and a steeper TTR slope as the text progresses, while LLM outputs tend toward greater lexical homogeneity [1].

Category 2 — Syntactic Complexity. Ten features characterise sentence-level structure: average sentence length, sentence length variance, clause density (approximated via comma frequency), subordination and coordination ratios, passive voice frequency (via a regex-based heuristic), a noun-phrase

proxy (mid-sentence capitalised tokens), verb density, mean relative word position within sentences, and sentence length skewness. Research on AI-generated text has consistently found that LLMs produce more uniform sentence lengths with lower variance [18], making these features informative even under paraphrase attacks.

Category 3 — Discourse Coherence. Eight features quantify text coherence at the sentence and paragraph level: mean Jaccard similarity between adjacent sentences (lexical cohesion), pronoun density, transition word frequency and diversity, paragraph-level cosine similarity between adjacent paragraphs, topic consistency between the first and last paragraphs, information flow regularity, and semantic repetition rate. Crucially, AI-generated text tends to exhibit high local cohesion (smooth adjacency transitions) combined with lower global topic consistency, a pattern that stylometric features can detect even when individual word choices are perturbed.

Category 4 — Style Fingerprinting. Seven features capture surface-level authorial fingerprints: punctuation density, capitalisation ratio, digit frequency, quote usage, parenthetical frequency, paragraph length variance, and list structure frequency. While these features are the easiest to alter through obfuscation, their combination with the more robust categories above provides useful complementary signal.

Category 5 — Perplexity-Based Signals. Five self-contained, LM-independent features approximate perplexity-related signals using corpus statistics. We compute forward and backward word-level entropy using the empirical unigram distribution of the text itself, their ratio, the burstiness of information (variance in per-word surprisal values), and mean surprisal. These features avoid the need for an external language model—a deliberate design choice that keeps inference lightweight and avoids overfitting to a specific generator.

The extracted 38-dimensional vector is projected to 128 dimensions through a small MLP:

$$h_{\text{style}} = \text{GELU}(\text{LayerNorm}(\mathbf{W}_s \mathbf{f}_{\text{style}} + \mathbf{b}_s)), \quad h_{\text{style}} \in \mathbb{R}^{128} \tag{1}$$

where $\mathbf{f}_{\text{style}} \in \mathbb{R}^{38}$ is the raw feature vector and $\mathbf{W}_s$, $\mathbf{b}_s$ are learned parameters.

3.3. Cross-Attention Fusion Layer

Naive concatenation of the semantic and stylometric representations would ignore the rich structural relationship between the two modalities. Instead, RADAR uses a multi-head cross-attention mechanism that allows the semantic representation to selectively query the stylometric representation, learning which stylometric dimensions are most informative given the specific semantic content.

Formally, the projected style vector $h_{\text{style}} \in \mathbb{R}^{128}$ is first linearly projected to match the semantic dimension:

$$\tilde{h}_{\text{style}} = \mathbf{W}_{s2e}\, h_{\text{style}}, \quad \tilde{h}_{\text{style}} \in \mathbb{R}^{1024} \tag{2}$$

Multi-head cross-attention is then applied with the semantic embedding as query and the projected style vector as key and value:

$$h_{\text{attended}} = \text{MHA}\left(\underbrace{h_{\text{semantic}}}_{\text{query}}, \underbrace{\tilde{h}_{\text{style}}}_{\text{key/value}}\right) + h_{\text{semantic}} \tag{3}$$

followed by layer normalisation. The attention uses 8 heads. A residual connection is applied to prevent information loss from the semantic branch.

Finally, the attended semantic representation is concatenated with the original projected style vector and passed through a fusion MLP:

$$h_{\text{fused}} = \text{GELU}(\text{LayerNorm}(\mathbf{W}_f\, [h_{\text{attended}};\, h_{\text{style}}] + \mathbf{b}_f)), \quad h_{\text{fused}} \in \mathbb{R}^{512} \tag{4}$$

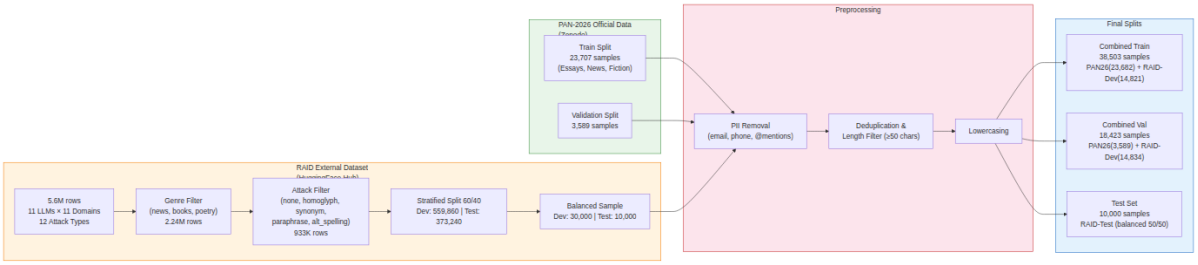


Figure 2: Overview of the data preparation pipeline. PAN-2026 official data and the RAID external dataset are preprocessed, filtered, and merged to form the final training, validation, and test splits.

This two-level fusion—cross-attention followed by concatenation—ensures that the classifier receives both the cross-modally attended information and the unmodified stylometric projection, preventing the attention mechanism from wholly discarding fine-grained stylometric detail.

3.4. UACC: Unified Authorship Classification Classifier

The fused representation $h_{\text{fused}} \in \mathbb{R}^{512}$ is passed through a single linear classifier to produce a scalar logit:

$$z = \mathbf{w}_c^\top h_{\text{fused}} + b_c, \quad z \in \mathbb{R} \tag{5}$$

The output confidence score is obtained via sigmoid activation:

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}, \quad \hat{y} \in [0, 1] \tag{6}$$

A score of $\hat{y} < 0.5$ indicates a human-authored text; $\hat{y} > 0.5$ indicates AI-authored text; and $\hat{y} = 0.5$ is treated as "undecidable" in the PAN evaluation framework.

3.5. Loss Function and Class Imbalance

Training uses binary cross-entropy with logits. The PAN-2026 training split contains approximately 61.7% AI-authored and 38.3% human-authored samples, introducing a class imbalance that can bias the model toward the majority class. To compensate, we apply a positive-class weight $w_+ = N_{\text{neg}}/N_{\text{pos}}$, where N_{neg} and N_{pos} are the number of human and AI samples respectively. For our combined training set this yields $w_+ \approx 0.746$, i.e. machine-authored samples are down-weighted relative to human-authored samples, encouraging the model to maintain balanced recall.

3.6. Parameter Count

The total parameter count is approximately 399.7 million, the vast majority of which resides in the ModernBERT-large encoder (395.4 million). The stylometric projection, fusion, and classification head account for approximately 4.9 million trainable parameters (1.2% of the total). This modular separation is exploited in our two-stage training curriculum described in Section 5.

4. Data

Our training pipeline combines two complementary data sources: the official PAN-2026 competition corpus and a curated subset of the RAID benchmark. Figure 2 provides a schematic overview of the data preparation process.

4.1. PAN-2026 Dataset

The official task corpus [3] was downloaded from Zenodo.[1] It consists of newline-delimited JSON files where each sample carries the fields `id`, `text`, `model`, `label` (0 = human, 1 = AI), and `genre`.

The training split comprises 23,707 samples across three genres: *essays* (diverse expository writing), *news* (articles from last year's PAN-2025 task with additional GPT-4o samples), and *fiction*. The validation split contains 3,589 samples drawn from the same distribution.

Label distribution. The training split is moderately imbalanced: 61.7% of samples are AI-authored and 38.3% are human-authored. The genre distribution is dominated by fiction (approximately 61.9% of training samples), with essays and news comprising the remainder.

AI generators. The machine-authored samples in the training set were produced by multiple models, including GPT-4o, Claude 3.5, Gemini 1.5 Pro, and Llama 3-based models. Importantly, the LLMs were instructed to mimic a specific human author's writing style, which substantially increases detection difficulty.

4.2. RAID External Dataset

To improve robustness to out-of-distribution generators and adversarial obfuscation, we augment training with a subset of the RAID benchmark [10]. RAID contains approximately 5.6 million samples generated by 11 LLMs across 11 domains and 12 adversarial attack types, making it the largest adversarially diverse AI-text benchmark currently available.

Genre filtering. We retain only three RAID domains that are semantically aligned with the PAN-2026 genres: *news* (↔ PAN news), *books* (↔ PAN fiction), and *poetry* (↔ PAN essays). After genre filtering, 2.24 million samples remain.

Attack filtering. We retain five attack types that reflect the kinds of obfuscation used in the PAN evaluation:

- `none` — clean AI/human baseline samples;
- `homoglyph` — character-level Unicode substitution (used in the PAN test set);
- `synonym` — lexical paraphrase via synonym replacement;
- `paraphrase` — full sentence-level rewriting (hardest attack type);
- `alternative_spelling` — region-based spelling variation.

After attack filtering, 933,100 samples remain.

Stratified split and sampling. The filtered RAID set is split 60/40 (stratified by label) into a *dev* pool (559,860 samples) and a held-out *test* set (373,240 samples). To keep training tractable on a single GPU and to enforce a balanced signal, we draw a stratified subsample from the dev pool: 30,000 samples with exactly 50% human and 50% AI. The balanced RAID test subsample (10,000 samples, 50/50) is used exclusively for out-of-domain evaluation. Attack types are uniformly represented in the subsample (approximately 6,000 samples per attack type in the dev sample).

[1] https://zenodo.org/records/14962653

Table 2
Final dataset statistics for each split.

Split	Total	Human	AI	Human %
Train (PAN26 + RAID)	38,503	16,449	22,054	42.7
Val (PAN26 + RAID)	18,423	8,569	9,854	46.5
Test (RAID only)	10,000	5,000	5,000	50.0
PAN26 Train	23,682	9,076	14,606	38.3
PAN26 Val	3,589	1,277	2,312	35.6
RAID Dev (subsample)	30,000	15,000	15,000	50.0
RAID Test (subsample)	10,000	5,000	5,000	50.0

4.3. Data Preprocessing

We apply a lightweight but consistent preprocessing pipeline to all splits:

1. **PII removal**: email addresses, phone numbers, and Twitter/social-media user mentions (`@username`) are replaced with anonymous tokens (`[EMAIL]`, `[PHONE]`, `[USER]`).
2. **Deduplication**: exact duplicate texts are removed.
3. **Length filter**: texts shorter than 50 characters are discarded.
4. **Lowercasing**: all text is converted to lower case to reduce sparsity in the stylometric feature space (tokenisation for the transformer is not affected, as the tokeniser handles casing internally).

After preprocessing, the PAN-2026 training split reduces from 23,707 to 23,682 samples (25 duplicates removed). All RAID samples pass the length filter as RAID texts are generally long-form.

4.4. Final Dataset Statistics

Table 2 summarises the final dataset composition. The combined training set contains 38,503 samples: 23,682 from PAN-2026 and 14,821 from the RAID dev subsample (after the remaining RAID dev portion is allocated to validation). The validation set (18,423 samples) combines the official PAN-2026 validation split with the remaining RAID dev samples. The test set (10,000 samples, balanced 50/50) consists entirely of the RAID test subsample and is used exclusively for cross-domain robustness evaluation.

5. Experimental Setup

We conduct four training experiments designed to study the impact of the encoder freezing strategy and the training data composition. All experiments share the same RADAR architecture and hyperparameter configuration; the only variables are which model weights are updated and which dataset is used for training.

5.1. Model Initialisation

In all experiments, the ModernBERT-large encoder is initialised with the pretrained weights from the official `answerdotai/ModernBERT-large` checkpoint on the HuggingFace Hub. The SLPE projection layer, the cross-attention fusion module, and the classifier are randomly initialised.

5.2. Two-Stage Training Curriculum

Stage 1 — Frozen Encoder (Exp. 1). In the first stage, the ModernBERT-large encoder weights are frozen entirely (using `requires_grad=False`). Only the SLPE projection, the cross-attention fusion layer, and the classifier head are trained. With the encoder frozen, the number of trainable parameters

Table 3
Training hyperparameters used in all experiments.

Hyperparameter	Value
Base encoder	ModernBERT-large
Max sequence length	512 tokens
Style dimension (d_s)	38
Style projection dim	128
Fusion dimension	512
Attention heads (fusion)	8
Dropout	0.1
Learning rate	2×10^{-5}
LR scheduler	Cosine decay
Warmup steps	3% of total steps
Optimizer	AdamW (fused)
β_1, β_2	0.9, 0.999
ε	1×10^{-8}
Train batch size	32
Eval batch size	64
Gradient accumulation	1
Epochs	10
Positive class weight	0.746
Seed	42

drops to approximately 4.9 million (1.2% of total), which substantially accelerates training and reduces GPU memory pressure.

This stage serves two purposes. First, it trains the stylometric and fusion modules to extract complementary information from the static encoder representations, ensuring that the style branch learns meaningful projections. Second, the resulting checkpoint provides a well-initialised starting point for Stage 2 that is considerably better than random initialisation of the non-encoder components.

Stage 2 — Full Fine-Tuning (Exp. 2). In the second stage, the Stage 1 checkpoint is loaded and all encoder weights are unfrozen. All 399.7 million parameters are jointly optimised. The lower learning rate relative to Stage 1 (both use 2×10^{-5} due to the small batch size and cosine scheduler) prevents catastrophic forgetting of the encoder's pre-trained representations while allowing targeted fine-tuning on the authorship signal.

5.3. Ablation Experiments

Two additional single-stage experiments isolate the contribution of each data source:

- **Exp. 3 — PAN-2026 Only (Frozen Encoder)**: Same as Stage 1 but trained exclusively on the PAN-2026 training set (23,682 samples). This measures in-domain performance and quantifies the marginal contribution of the RAID augmentation.
- **Exp. 4 — RAID Only (Frozen Encoder)**: Same as Stage 1 but trained exclusively on the RAID dev subsample (30,000 samples). This measures out-of-domain generalisation and the effect of adversarial attack diversity.

5.4. Hyperparameters

Table 3 lists the training hyperparameters used across all experiments. All experiments were trained on a single NVIDIA A100 40 GB GPU using 32-bit mixed precision with `bfloat16` for the encoder and full precision for the fusion/classification head.

5.5. Evaluation Protocol

All models are evaluated with the official PAN-2026 evaluator (`pan26-evaluator` [24]), which computes the following metrics:

- **ROC-AUC**: area under the receiver operating characteristic curve.
- **Brier**: complement of the Brier score ($1 - \text{MSE}$).
- **C@1**: modified accuracy that assigns non-answers (score $= 0.5$) the average accuracy of the remaining cases, rewarding abstention over wrong answers.
- **F1**: harmonic mean of precision and recall.
- **F0.5u**: precision-weighted F-measure that treats non-answers as false negatives.
- **Mean**: arithmetic mean of all five metrics above.

We report evaluation results on three sets: the combined validation set (Val, 18,423 samples), the PAN-2026 validation set only (PAN26-Val, 3,589 samples), and the RAID test set (RAID-Test, 10,000 samples). The last two allow us to separately assess in-domain and out-of-domain performance.

5.6. Baseline Comparison

We compare our results against three published baselines provided by the organisers:

1. **TF-IDF SVM**: linear SVM trained on TF-IDF bag-of-words features.
2. **PPMd Compression Cosine** [12, 13]: unsupervised compression-based authorship similarity model.
3. **Binoculars** [8]: zero-shot detector using cross-model perplexity divergence.

Note that the official baseline numbers were computed on the *pure PAN-2026 validation set* (3,589 samples). Our combined validation set is larger (18,423 samples) and includes harder adversarial examples from RAID. Direct numerical comparison should therefore be interpreted with this caveat in mind; we use the PAN26-Val subset for a fair in-domain comparison.

6. Results and Discussion

This section presents and analyses the results of our four experiments alongside the official task baselines.

6.1. Main Results

Table 4 presents the evaluation results for all four RADAR configurations on the combined validation set and on the RAID test set. Table 5 reports performance on the PAN-2026 validation set in isolation, allowing a direct comparison with the published baseline numbers.

6.2. Analysis

Two-stage training yields the best overall performance. The end-to-end fine-tuned model (Exp. 2) achieves the strongest results across all metrics and on all evaluation sets. On the PAN-2026 validation set, it reaches near-perfect performance (Mean = 0.997, ROC-AUC = 1.000), substantially outperforming the TF-IDF SVM baseline (Mean = 0.978) despite the latter being a strong and well-calibrated supervised model. On the harder combined validation set that includes adversarial RAID examples, Exp. 2 achieves Mean = 0.978, matching the SVM's in-domain performance on a considerably more difficult evaluation distribution.

Table 4
Results on the combined validation set (Val, 18,423 samples) and the RAID test set (RAID-Test, 10,000 samples). Best results per metric are **bolded**.

System	Split	ROC-AUC	Brier	C@1	F1	F0.5u	Mean
Exp. 1: Enc.-Freeze (PAN26+RAID)	Val	0.976	0.940	0.919	0.921	0.946	0.941
Exp. 1: Enc.-Freeze (PAN26+RAID)	RAID-Test	0.970	0.934	0.913	0.907	0.941	0.933
Exp. 2: End-to-End (PAN26+RAID)	Val	**0.992**	**0.973**	**0.971**	**0.973**	**0.981**	**0.978**
Exp. 2: End-to-End (PAN26+RAID)	RAID-Test	**0.990**	**0.967**	**0.964**	**0.964**	**0.974**	**0.972**
Exp. 3: Enc.-Freeze (PAN26 only)	Val	—	—	—	—	—	—
Exp. 3: Enc.-Freeze (PAN26 only)	RAID-Test	0.716	0.731	0.650	0.659	0.649	0.681
Exp. 4: Enc.-Freeze (RAID only)	Val	0.972	0.937	0.914	0.913	0.929	0.933
Exp. 4: Enc.-Freeze (RAID only)	RAID-Test	0.974	0.941	0.920	0.918	0.935	0.938

Table 5
Comparison with official task baselines on the PAN-2026 validation set (3,589 samples). Baselines are evaluated in-domain; RADAR results on this same split are shown for direct comparison.

System	ROC-AUC	Brier	C@1	F1	F0.5u	Mean
TF-IDF SVM	0.996	0.951	0.984	0.980	0.981	0.978
PPMd CBC	0.786	0.799	0.757	0.812	0.778	0.786
Binoculars	0.918	0.867	0.844	0.872	0.882	0.877
Exp. 1 (Enc.-Freeze)	0.997	0.982	0.976	0.982	0.978	0.983
Exp. 2 (End-to-End)	**1.000**	**0.997**	**0.996**	**0.997**	**0.996**	**0.997**
Exp. 3 (PAN26 only)	0.997	0.983	0.979	0.984	0.982	0.985
Exp. 4 (RAID only)	0.795	0.651	0.644	0.784	0.694	0.714

Frozen encoder stage provides a strong intermediate. Experiment 1 (frozen encoder, combined data) achieves Mean = 0.941 on the combined validation set and Mean = 0.933 on the RAID test set. These results confirm that the SLPE and fusion modules already extract substantial discriminative signal from the frozen encoder representations combined with stylometric features—even with only 1.2% of parameters being updated. The Stage 1 checkpoint provides a well-initialised starting point that accelerates Stage 2 convergence.

Data composition critically affects cross-domain robustness. Experiment 3 (trained on PAN-2026 only) achieves excellent in-domain performance (Mean = 0.985 on PAN26-Val) but catastrophically low performance on the RAID test set (Mean = 0.681), dropping over 30 percentage points. This domain gap illustrates the danger of fitting exclusively to the in-domain distribution when the evaluation set contains new generators and adversarial obfuscations.

Conversely, Experiment 4 (trained on RAID only) is quite robust on the RAID test set (Mean = 0.938) but degrades significantly on the PAN-2026 in-domain validation (Mean = 0.714). This suggests that RAID alone is not representative enough to fully cover the stylistic characteristics of PAN-2026 texts.

The full combination of both datasets (Exp. 1 and Exp. 2) achieves a balanced and superior performance profile, confirming that multi-source heterogeneous training is essential for genuine robustness.

RADAR outperforms zero-shot baselines by a substantial margin. Even the frozen-encoder Stage 1 model (Exp. 1) substantially outperforms both Binoculars (Mean = 0.877) and PPMd CBC (Mean = 0.786) on the PAN-2026 validation set (Exp. 1: Mean = 0.983). This confirms that training a hybrid model that combines semantic and stylometric signals is more effective than unsupervised statistical approaches, even without full encoder fine-tuning.

Table 6
Selected training checkpoints for Experiment 1 (frozen encoder, combined data), showing key validation metrics at representative epochs.

Step	Epoch	Train Loss	Val Loss	ROC-AUC	C@1	F1	Mean
500	0.41	0.398	0.326	0.900	0.816	0.806	0.854
2500	2.08	0.242	0.217	0.961	0.885	0.883	0.915
5000	4.15	0.090	0.186	0.971	0.903	0.902	0.930
8000	6.64	0.079	0.164	0.976	0.919	0.922	0.940
11500	9.55	0.051	0.164	0.976	0.918	0.920	0.940
12000	9.97	0.038	0.163	0.976	0.918	0.920	0.940

Table 7
Selected training checkpoints for Experiment 2 (end-to-end fine-tuning), showing key validation metrics at representative epochs.

Step	Epoch	Train Loss	Val Loss	ROC-AUC	C@1	F1	Mean
500	0.41	0.210	0.181	0.982	0.904	0.902	0.934
2000	1.66	0.048	0.087	0.993	0.961	0.962	0.973
4000	3.32	0.035	0.136	0.989	0.955	0.957	0.967
8500	7.06	0.013	0.146	0.991	0.962	0.964	0.972
11000	9.14	0.011	0.135	0.992	0.971	0.973	0.978
12000	9.97	0.003	0.135	0.992	0.971	0.973	0.978

Value of stylometric features. An implicit ablation is provided by contrasting Exp. 3 (PAN-2026 only, frozen encoder) against the TF-IDF SVM. While both achieve comparable in-domain performance (Mean 0.983 vs 0.978), RADAR's advantage emerges clearly under distribution shift. The TF-IDF SVM is purely lexical and known to fail under synonym substitution and paraphrase; the RADAR architecture's stylometric features are more stable under these perturbations because they capture structural patterns rather than surface vocabulary.

Calibration. The Brier score, which measures probabilistic calibration, consistently improves from Stage 1 to Stage 2. The end-to-end model achieves Brier = 0.997 on PAN26-Val and Brier = 0.967 on the RAID test set, indicating that the confidence scores are well-calibrated and reliable for downstream use—important for the "undecidable" abstention mechanism in the PAN evaluation.

6.3. Training Dynamics

Tables 6 and 7 present selected checkpoint metrics for Experiments 1 and 2. In Stage 1 (frozen encoder), the training loss decreases from 0.398 to 0.038 over 10 epochs, while the validation loss stabilises around 0.163, with the best checkpoint at epoch 9.5 (Mean = 0.941). In Stage 2 (full fine-tuning), the training loss decreases substantially faster from 0.210 to 0.003, reflecting the additional capacity of the unfrozen encoder, while the validation loss reaches its minimum around epoch 9 before showing minor overfitting signs. We select the best checkpoint by Mean score on the validation set.

7. Conclusion

This paper presented RADAR, a hybrid neural architecture for robust AI-generated text detection that combines deep contextual semantics with handcrafted stylometric signals. Built on ModernBERT-large, RADAR couples a frozen or fine-tuned encoder (ARSE) with a stylometric feature extractor (SLPE) and a cross-attention fusion module that lets semantic and stylometric representations interact in a principled way before a single binary classifier produces a calibrated confidence score.

We trained RADAR through a two-stage curriculum on a heterogeneous dataset combining the official PAN-2026 training data and a balanced subsample of the RAID benchmark, which encompasses a wide variety of AI generators and text-level adversarial perturbations. The final end-to-end model achieves Mean = 0.997 on the PAN-2026 validation set, surpassing all published baselines, and Mean = 0.972 on the RAID test set, demonstrating strong cross-domain generalisation.

The ablation experiments reveal several key findings. First, multi-source training is indispensable for robust performance: models trained on a single source degrade substantially under domain shift, with Exp. 3 (PAN-2026 only) dropping from Mean = 0.985 in-domain to Mean = 0.681 on RAID. Second, the two-stage curriculum consistently outperforms single-stage training: unfreezing the encoder only after the non-encoder modules have converged leads to stable and fast convergence of the full model. Third, stylometric features complement semantic representations in a way that improves robustness to adversarial obfuscation attacks such as synonym substitution and homoglyph injection, which systematically fool purely lexical models.

Limitations. Although RADAR achieves strong performance on the evaluated benchmarks, several limitations warrant caution in deployment. The model is trained predominantly on English texts; its effectiveness on other languages or mixed-language content is unknown. The architecture's maximum input length of 512 tokens means that the signal from longer documents is truncated. Furthermore, while RAID covers a diverse range of adversarial attacks, new obfuscation strategies may emerge that are not yet represented in the training distribution. Finally, the stylometric features are extracted via deterministic rules; learned stylometric representations [13] could potentially offer richer signals.

Future Work. Promising directions for future work include: (i) extending the evaluation to multilingual texts and multilingual base encoders; (ii) incorporating longer-range context through hierarchical encoding or sparse attention; (iii) exploring adversarial training with additional obfuscation types at training time; and (iv) applying uncertainty calibration techniques to further improve the reliability of confidence scores for human-in-the-loop review workflows.

In summary, RADAR demonstrates that combining pre-trained language model representations with lightweight, theoretically motivated stylometric signals through a principled cross-attention fusion mechanism is a highly effective and generalisable approach to AI-generated text detection in adversarial conditions.

Acknowledgments

The author thanks the PAN 2026 organisers for providing the task infrastructure and the TIRA evaluation platform [24]. We are grateful to the creators of the RAID benchmark [10] for making their dataset publicly available.

Declaration on Generative AI

The author(s) have not employed any Generative AI tools in the preparation of this work.

References

[1] M. Jakesch, J. Hancock, M. Naaman, Human Heuristics for AI-Generated Language Are Flawed, in: Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems, ACM, 2023. doi:10.1145/3544548.3581318.

[2] J. Bevendorff, D. Dementieva, M. Fröbe, B. Gipp, A. Greiner-Petter, J. Karlgren, M. Mayerl, P. Nakov, A. Panchenko, M. Potthast, A. Shelmanov, E. Stamatatos, B. Stein, Y. Wang, M. Wiegmann, E. Zangerle, Overview of PAN 2025: Generative AI Authorship Verification, Multi-Author Writing

Style Analysis, Multilingual Text Detoxification, and Generative Plagiarism Detection, in: Experimental IR Meets Multilinguality, Multimodality, and Interaction. Proceedings of the Fourteenth International Conference of the CLEF Association (CLEF 2025), Lecture Notes in Computer Science, Springer, Berlin Heidelberg New York, 2025.

[3] J. Bevendorff, D. Dementieva, M. Fröbe, B. Gipp, A. Greiner-Petter, J. Karlgren, M. Mayerl, P. Nakov, A. Panchenko, M. Potthast, A. Shelmanov, E. Stamatatos, B. Stein, Y. Wang, M. Wiegmann, E. Zangerle, Overview of PAN 2026: Voight-Kampff Generative AI Detection, Multi-Author Writing Style Analysis, and Multilingual Text Detoxification, in: Working Notes of CLEF 2026, CEUR Workshop Proceedings, CEUR-WS.org, 2026.

[4] J. Bevendorff, I. Borenstein, J. Kalo, E. Llovejoy, M. Mayerl, M. Potthast, B. Stein, M. Wiegmann, Generative AI Authorship Verification in the Voight-Kampff Game, in: Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics, Association for Computational Linguistics, 2024, pp. 2025–2041.

[5] J. Bevendorff, M. Mayerl, M. Potthast, B. Stein, Overview of the PAN 2025 Generative AI Detection Task, in: Working Notes of CLEF 2025, CEUR Workshop Proceedings, CEUR-WS.org, 2025.

[6] E. Mitchell, Y. Lee, A. Khazatsky, C. D. Manning, C. Finn, DetectGPT: Zero-Shot Machine-Generated Text Detection using Probability Curvature, arXiv preprint arXiv:2301.11305 (2023). URL: https://arxiv.org/abs/2301.11305.

[7] G. Bao, Y. Zhao, Z. Teng, L. Yang, Y. Zhang, Fast-DetectGPT: Efficient Zero-Shot Detection of Machine-Generated Text via Conditional Probability Curvature, arXiv preprint arXiv:2305.18149 (2023). URL: https://arxiv.org/abs/2305.18149.

[8] A. Hans, A. Schwarzschild, V. Cherepanova, H. Kazemi, A. Saha, M. Goldblum, J. Geiping, T. Goldstein, Binoculars: Zero-Shot Detection of LLM-Generated Text, in: Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics, Association for Computational Linguistics, 2024, pp. 1235–1247.

[9] J. Su, T. Y. Zhuo, D. Wang, P. Nakov, DetectLLM: Leveraging Log Rank Information for Zero-Shot Detection of Machine-Generated Text, arXiv preprint arXiv:2310.05842 (2023). URL: https://arxiv.org/abs/2310.05842.

[10] L. Dugan, A. Hwang, F. Trhlík, A. Zhu, G. Durrett, C. Callison-Burch, RAID: A Shared Benchmark for Robust Evaluation of Machine-Generated Text Detectors, in: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics, Association for Computational Linguistics, 2024, pp. 14085–14109.

[11] M. Koppel, J. Schler, Authorship Verification as a One-Class Classification Problem, in: Proceedings of the 20th International Conference on Machine Learning (ICML 2004), 2004.

[12] D. Sculley, C. E. Brodley, Compression and Machine Learning: A New Perspective on Feature Space Vectors, in: Proceedings of the 29th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, ACM, 2006, pp. 684–685. doi:10.1145/1148170.1148307.

[13] O. Halvani, C. Winter, L. Graner, Authorship Verification Based on Compression-Models, in: Proceedings of the 12th International Conference on Availability, Reliability and Security, ACM, 2017. doi:10.1145/3098954.3104050.

[14] V. S. Sadasivan, A. Kumar, S. Balasubramanian, W. Wang, S. Feizi, Can AI-Generated Text be Reliably Detected?, arXiv preprint arXiv:2303.11156 (2023). URL: https://arxiv.org/abs/2303.11156.

[15] D. Ippolito, D. Duckworth, C. Callison-Burch, D. Eck, Automatic Detection of Generated Text is Easiest when Humans are Fooled, in: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, Association for Computational Linguistics, 2020, pp. 1529–1541.

[16] A. Uchendu, Z. Ma, T. Le, R. Zhang, D. Lee, TURINGBENCH: A Benchmark Environment for Turing Test in the Age of Neural Text Generation, in: Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, 2021, pp. 3311–3326.

[17] J. Bevendorff, E. Stamatatos, M. Potthast, B. Stein, Overview of the Voight-Kampff Generative AI Authorship Verification Task at PAN and ELOQUENT 2024, in: Proceedings of the 7th Workshop

on Challenges and Applications of Automated Extraction of Socio-political Events from Text (CASE 2024), Association for Computational Linguistics, 2024.

[18] T. Schuster, R. Schuster, D. J. Shah, R. Barzilay, The Limitations of Stylometry for Detecting Machine-Generated Fake News, in: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), Association for Computational Linguistics, 2020, pp. 7022–7032.

[19] A. Uchendu, T. Le, K. Shu, D. Lee, Authorship Attribution for Neural Text Generation, in: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), Association for Computational Linguistics, 2020, pp. 8422–8432.

[20] J. Bevendorff, B. Stein, M. Hagen, M. Potthast, Generalizing Unmasking for Short Texts, in: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, Association for Computational Linguistics, 2019, pp. 4427–4437.

[21] B. Warner, A. Chaffin, B. Clavie, O. Weller, O. Hallström, S. Taghadouini, A. Galalgher, R. Biswas, F. Ladhak, T. Aarsen, N. Cooper, G. Adams, J. Howard, I. Poli, Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference, arXiv preprint arXiv:2412.13663 (2024). URL: https://arxiv.org/abs/2412.13663.

[22] E. Brunet, Le vocabulaire de Jean Giraudoux: Structure et évolution, Nice: Champion-Slatkine (1978).

[23] A. Honore, Some Simple Measures of Richness of Vocabulary, Association for Literary and Linguistic Computing Bulletin 7 (1979) 431–435.

[24] M. Fröbe, M. Wiegmann, N. Kolyada, B. Grahm, T. Elstner, F. Loebe, M. Hagen, B. Stein, M. Potthast, Continuous Integration for Reproducible Shared Tasks with TIRA.io, in: J. Kamps, L. Goeuriot, F. Crestani, M. Maistro, H. Joho, B. Davis, C. Gurrin, U. Kruschwitz, A. Caputo (Eds.), Advances in Information Retrieval. 45th European Conference on IR Research (ECIR 2023), Lecture Notes in Computer Science, Springer, Berlin Heidelberg New York, 2023, pp. 236–241. URL: https://link.springer.com/chapter/10.1007/978-3-031-28241-6_20. doi:10.1007/978-3-031-28241-6_20.